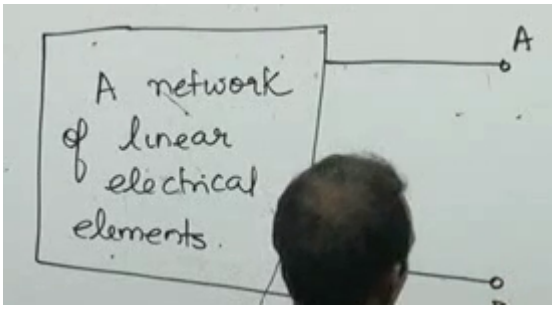
Session -5

Introduction to Thevenin's Theorem

Conceptualizing the Electrical Network

A complex network of linear electrical elements can be represented as a black box with two output terminals, A and B. This network is intended to perform a specific function and should not be altered.



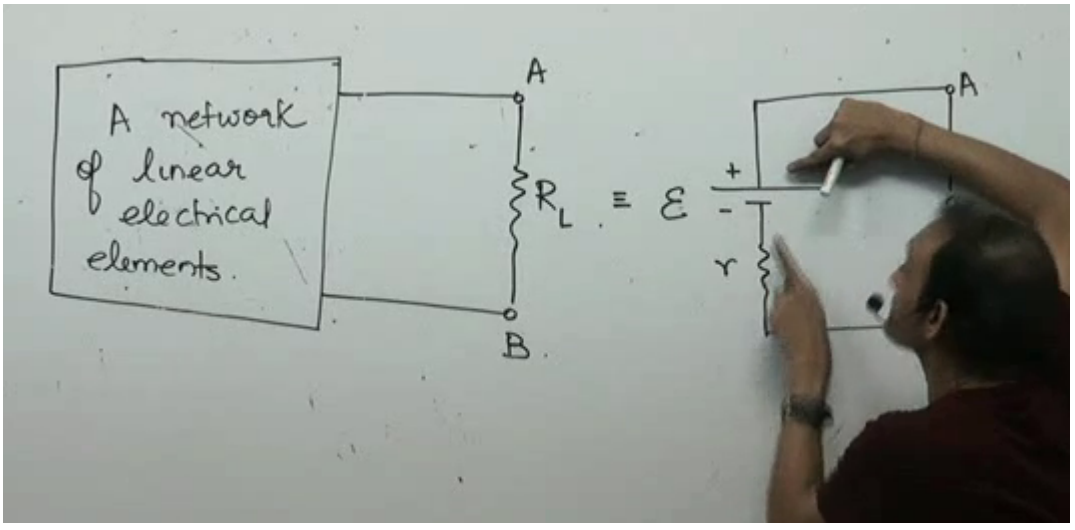
Electrical Network Block Diagram (Initial state, 0:59)

The Challenge of Load Calculation

Calculating the current through a variable load R_L connected across terminals A and B using standard methods like KVL and KCL for every change in load requires significant effort.

Thevenin's Theorem Reduction

Thevenin's theorem allows for the reduction of the entire active electrical network to a simpler equivalent circuit consisting of a battery with EMF ϵ and an internal resistance r connected in series across terminals A and B.



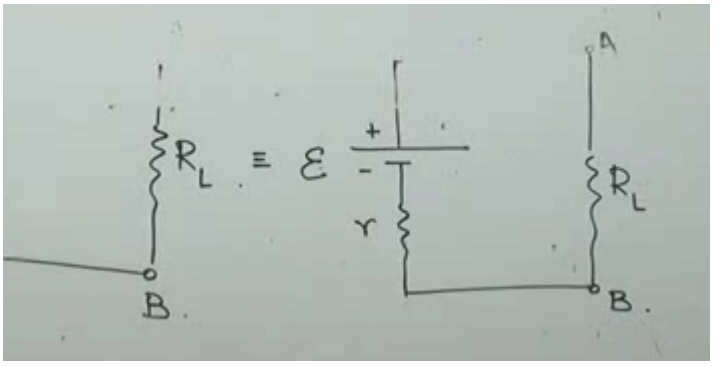
Thevenin equivalent circuit (Completed state, 3:44)

Calculating Current with the Equivalent Circuit

Once the network is reduced to its Thevenin equivalent, the current I through any new load R_L can be calculated using the formula:

$$I = \frac{\epsilon}{r + R_L}$$

where ϵ is the Thevenin EMF and r is the Thevenin equivalent resistance.

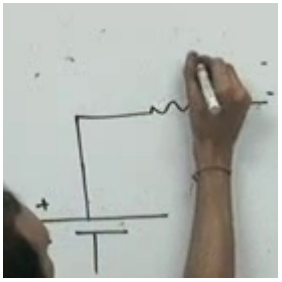


Electrical circuit equivalence diagram (Intermediate state, 3:56)

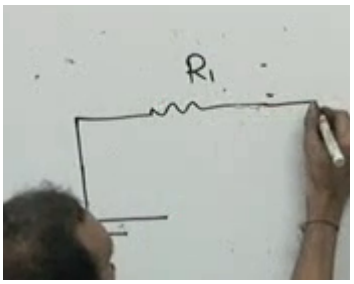
Thevenin's Theorem and Circuit Reduction

Active Electrical Network and Load

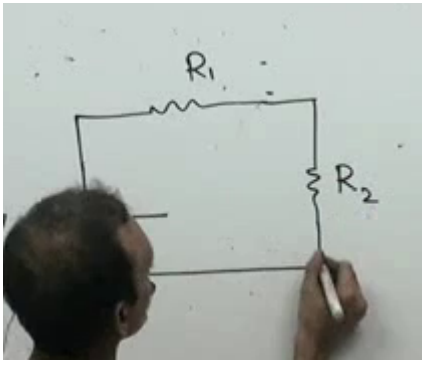
An active electrical network consists of voltage sources (e.g., V) and resistors (e.g., R_1, R_2, R_3). A load R is connected across terminals A and B. While the network remains constant, the load R may vary. Applying Kirchhoff's Voltage Law (KVL) to solve for the current through R for every change in load is cumbersome, especially as the number of loops increases.



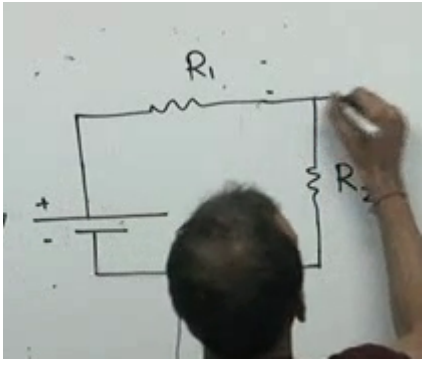
Simple Series Circuit Diagram



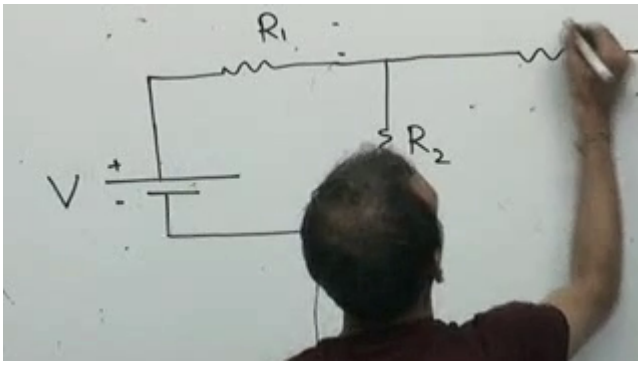
Partial Electrical Circuit



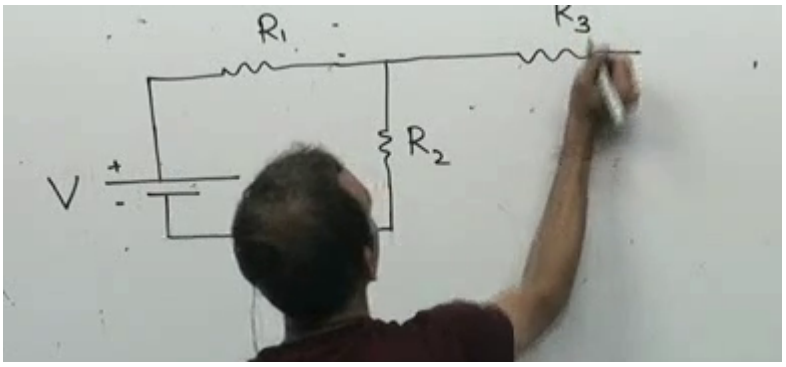
Series Circuit Diagram



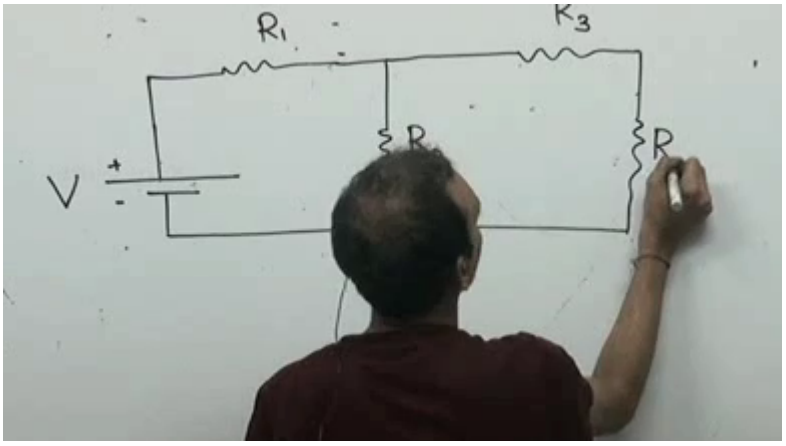
Series Circuit Diagram (Initial state, 4:38)



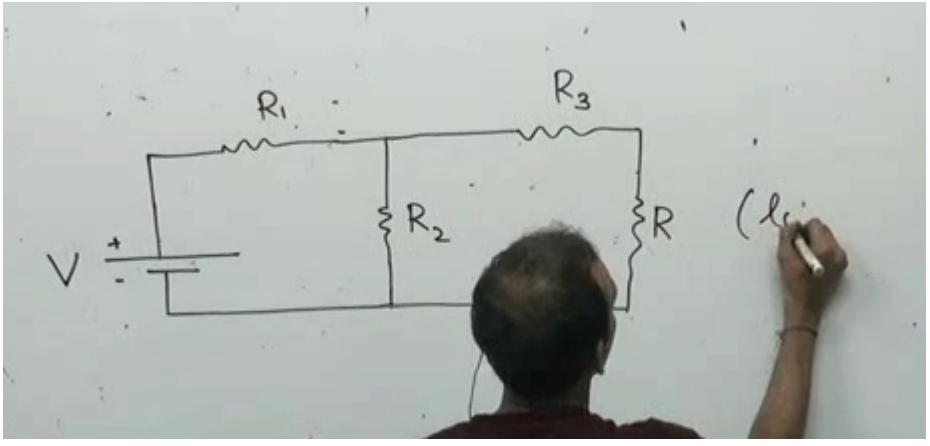
Electrical Circuit Schematic



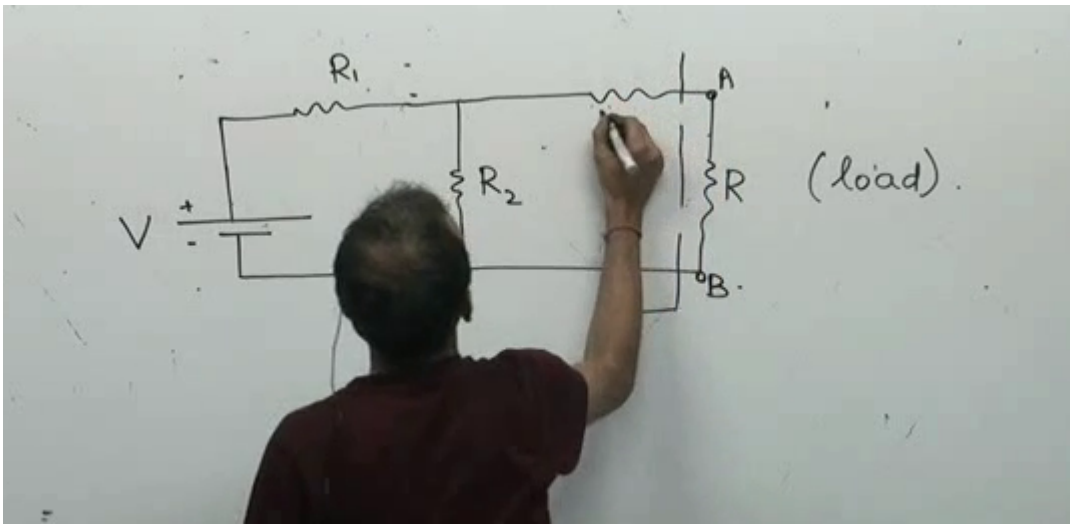
Electrical Circuit Diagram (Completed state, 4:42)



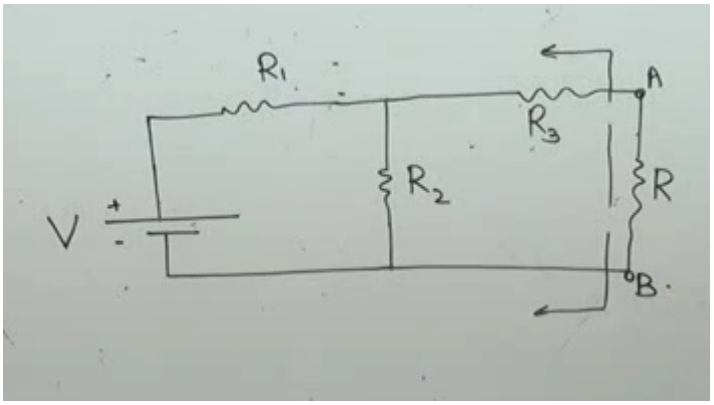
Resistive Circuit Schematic



Electrical circuit diagram



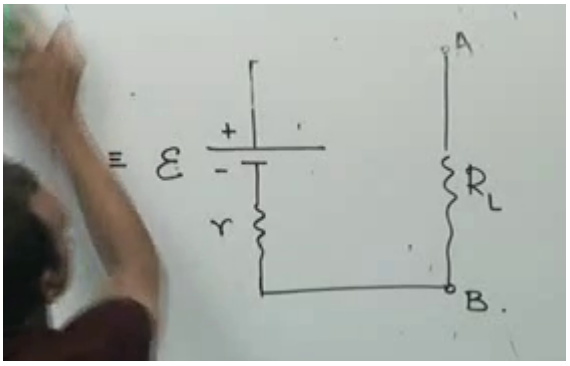
Electrical circuit diagram (Initial state, 5:05)



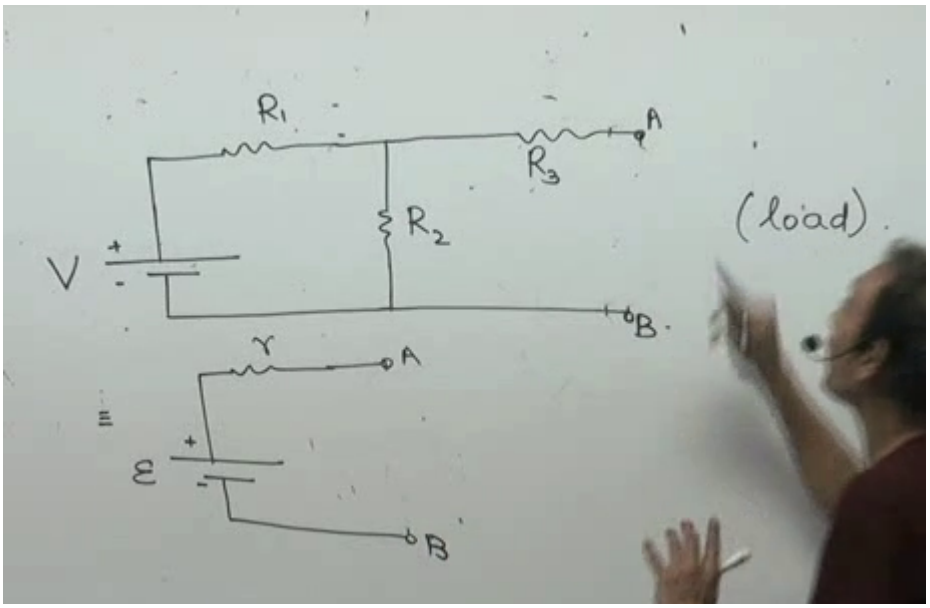
Electrical circuit diagram (Intermediate state, 5:16)

Thevenin Equivalent Circuit

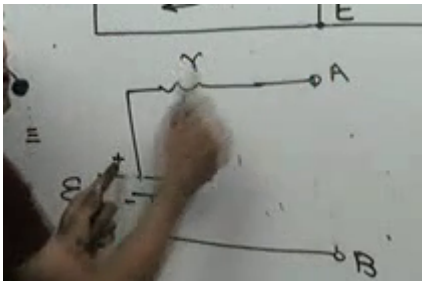
Thevenin's theorem allows reducing an active electrical network to an equivalent circuit consisting of an electromotive force (EMF), denoted as ϵ (or V_{th}), in series with an internal resistance r (or R_{th}), connected across terminals A and B. This equivalent circuit maintains the same behavior at terminals A and B regardless of the load connected.



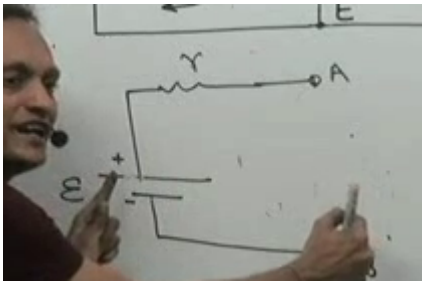
Electrical circuit diagram



Electrical circuit and its Thevenin equivalent (Intermediate state, 7:57)



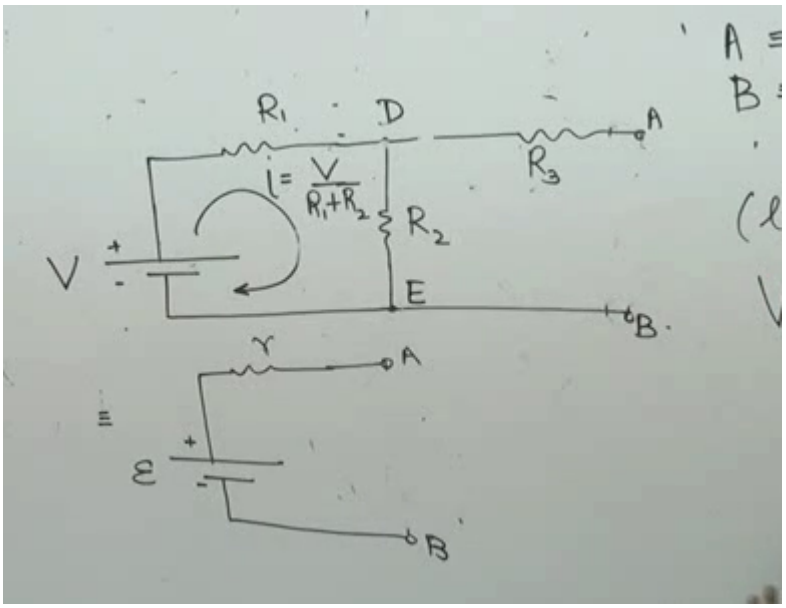
Simplified circuit diagram with internal resistance



Simplified circuit diagram of a voltage source with internal resistance

Calculating Thevenin Voltage (ϵ)

To find the Thevenin voltage ϵ, imagine an open circuit across terminals A and B by disconnecting the load. In this state, there is no current flowing through the load, but a potential difference exists between A and B. This open-circuit potential difference is equal to ϵ.



Thevenin Equivalent Circuit Derivation (Intermediate state, 10:37)

Worked Example: Determining ϵ

Given a circuit with a voltage source V and resistors R_1, R_2, R_3 in a loop, find the potential difference between A and B in an open-circuit state.

1. Calculate the current i through the loop: $i = \frac{V}{R_1+R_2}$.

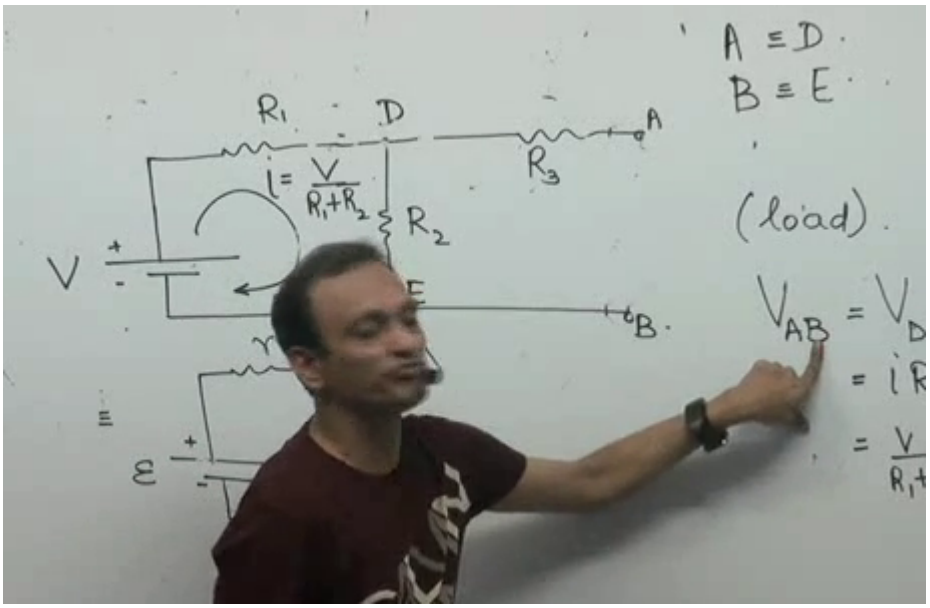
2. Analyze the potential at nodes A, B, D, and E. Since the circuit is open at A and B, there is no current in R_3, meaning there is no potential drop across R_3. Thus, potential at A equals potential at D ($V_A = V_D$).

3. Similarly, potential at B equals potential at E ($V_B = V_E$).

4. The potential difference V_{AB} is equal to the potential difference V_{DE}.

5. The potential drop across R_2 is $V_{DE} = i \cdot R_2$.

6. Substituting the current: $V_{AB} = \epsilon = \frac{V}{R_1+R_2} \cdot R_2$.

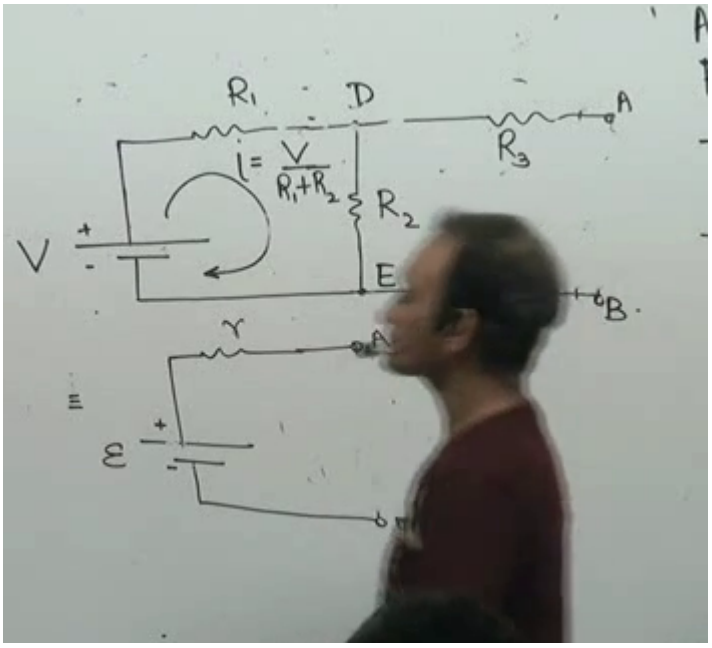


Voltage Divider Circuit Analysis (Completed state, 11:32)

Procedure for Finding Thevenin's Resistance (Rth)

Short-Circuiting EMF Sources

To determine the Thevenin resistance (r), first short-circuit all sources of electromotive force (EMF). Short-circuiting a source means connecting its terminals such that the potential difference across the source becomes 0. If the source has internal resistance, that resistance remains in the circuit.



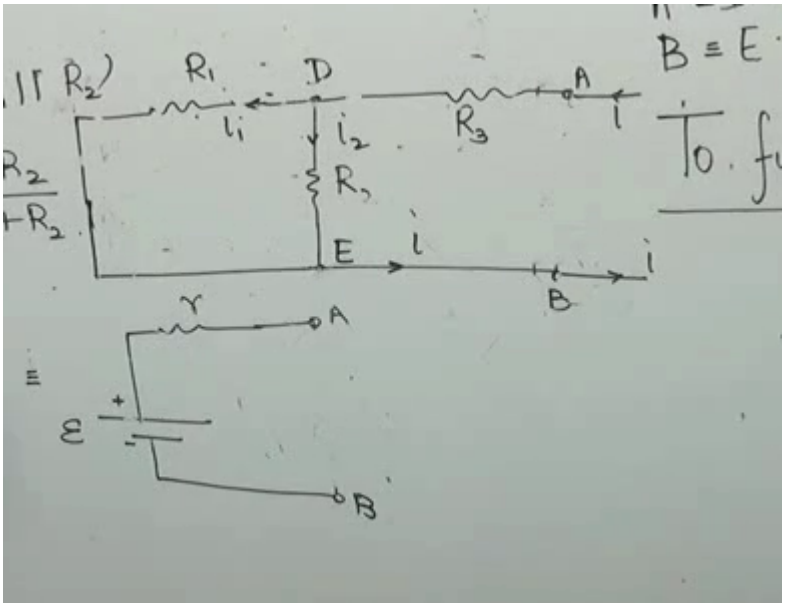
Thevenin Equivalent Circuit Derivation (Initial state, 12:20)

Determining Total Resistance

Imagine a current I entering terminal A and leaving terminal B. The total resistance encountered by this current is the Thevenin resistance (r).

By analyzing the circuit, current I first encounters resistor R_3. At node D, the current splits into I_1 (flowing through R_1) and I_2 (flowing through R_2). The currents I_1 and I_2 merge at node E. Because nodes E and F are electrically the same, resistors R_1 and R_2 are in parallel.

The total resistance r is the series combination of R_3 and the parallel combination of R_1 and R_2.



Circuit Diagram and Equivalent Circuit (Completed state, 16:34)

Calculation of Thevenin Resistance

The parallel resistance of R_1 and R_2 is given by $\frac{R_1 R_2}{R_1+R_2}$.

The total resistance r is the sum of R_3 and the parallel equivalent:

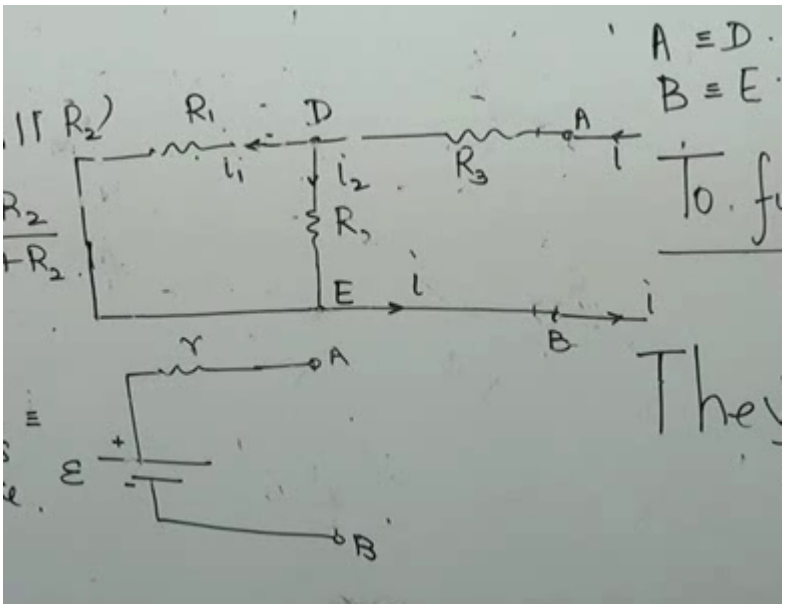
$$r = R_3 + \frac{R_1 R_2}{R_1 + R_2}$$

Conceptual Overview of Thevenin's Theorem

Definition and Purpose

Thevenin's theorem is a method used to simplify a complex electrical network into a simpler equivalent circuit consisting of a single voltage source (Thevenin's voltage, ϵ) and a single series resistance (Thevenin's resistance, r).

This process is referred to as 'Thevenizing' the circuit. The primary merit of this theorem is that the calculation of ϵ and r is a one-time effort. Once the network to the left of terminals A and B is replaced by this equivalent, the current i through any load R connected across A and B can be calculated using the formula $i = \frac{\epsilon}{R+r}$.



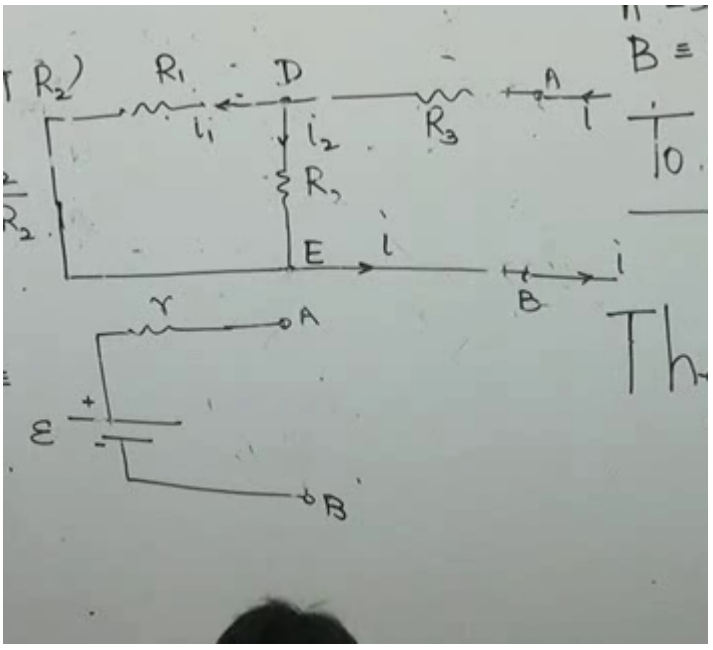
Thevenin Equivalent Circuit Derivation (Initial state, 18:31)

Handling Circuit Complexity

All loops within a network must be considered because each contributes to the values of ϵ and r. Even in complex networks with many loops, the calculation remains a one-time effort. The current flowing through the load depends on the load resistance R, but the Thevenin equivalent parameters (ϵ and r) are independent of the load connected across terminals A and B.

Load Connections and Network Tapping

Electrical networks are designed with standardized tapping points (sockets) A and B for connecting loads. If multiple loads are connected across these points, they are treated as being in parallel, effectively reducing to a single equivalent load resistance. If the tapping points A and B are changed, a fresh calculation for the new terminals is required to determine the new ϵ and r.



Circuit Analysis and Thevenin Equivalent (Completed state, 21:50)

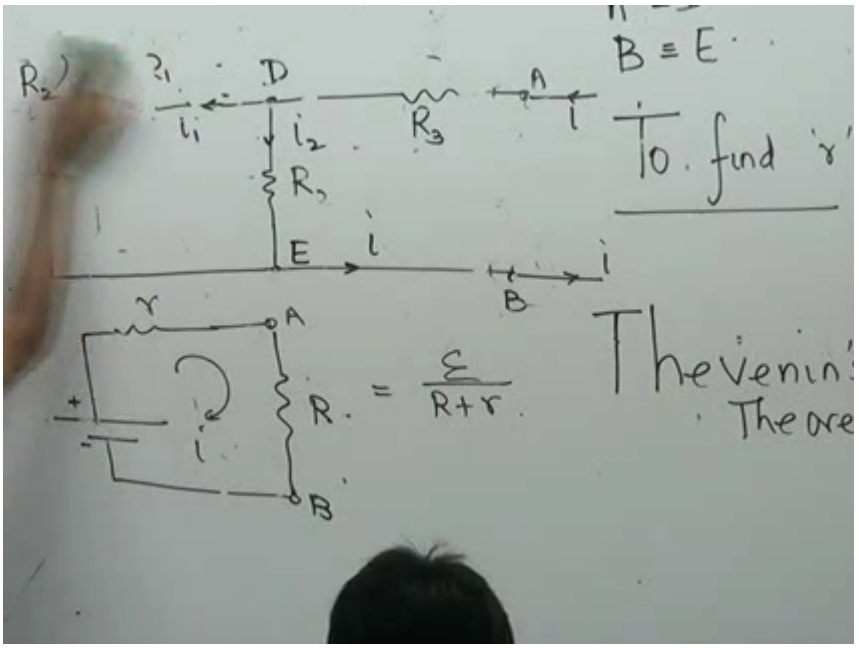
Linearity and Device Types

Thevenin's theorem applies to linear devices, such as resistors, capacitors, and inductors, which exhibit a linear relationship between voltage and current. Nonlinear devices are excluded from this scope.

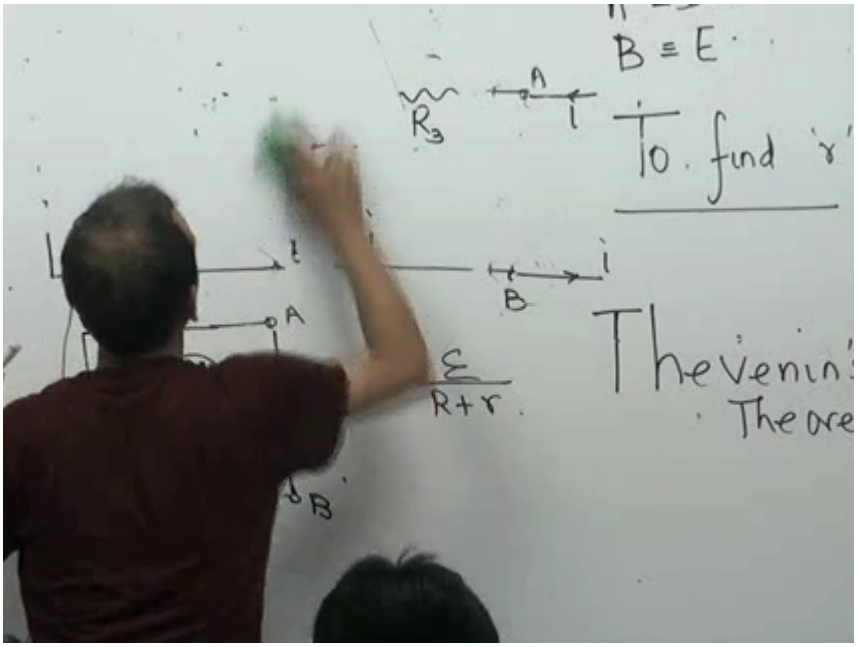
Thevenin's Theorem Circuit Analysis

Circuit Setup and Objective

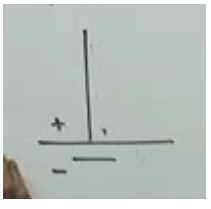
The circuit consists of a 10 V battery, a 3 Ω resistor (R1), a 7 Ω resistor, and a 5 Ω resistor. The objective is to 'Theveninize' the circuit to the left of terminals A and B, reducing it to a simpler equivalent network consisting of a Thevenin voltage (EPSA) and a Thevenin resistance (r) to which a variable load R can be connected.



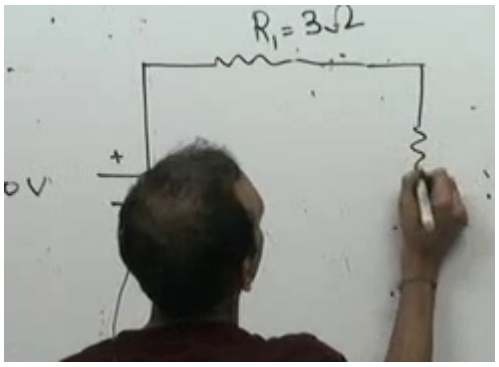
Electrical circuit analysis and Thevenin's Theorem (Initial state, 22:27)



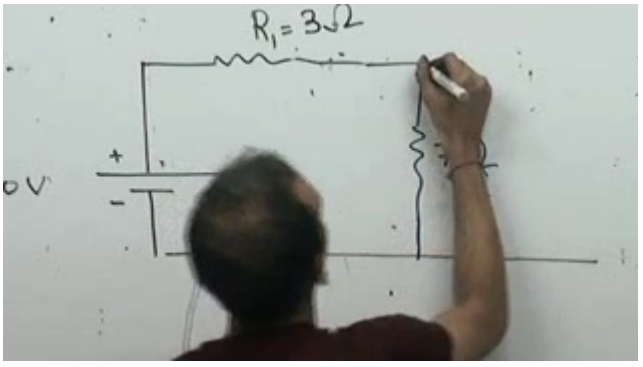
Electrical circuit diagram and Thevenin's Theorem (Completed state, 22:29)



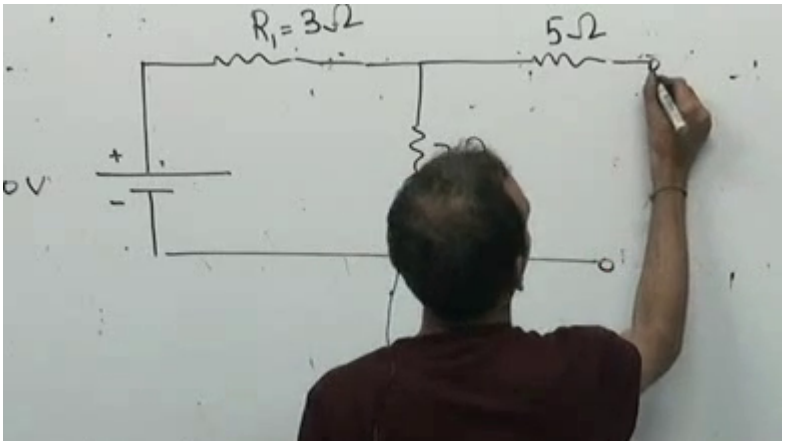
DC Voltage Source Symbol



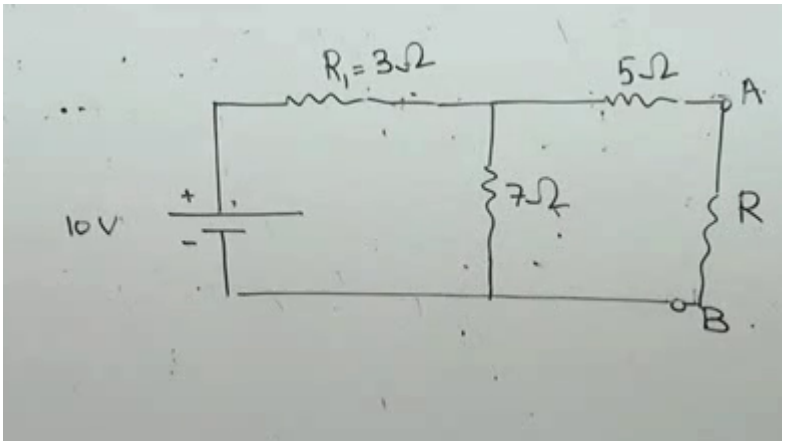
Series Circuit Diagram



Circuit Diagram (Initial state, 23:04)



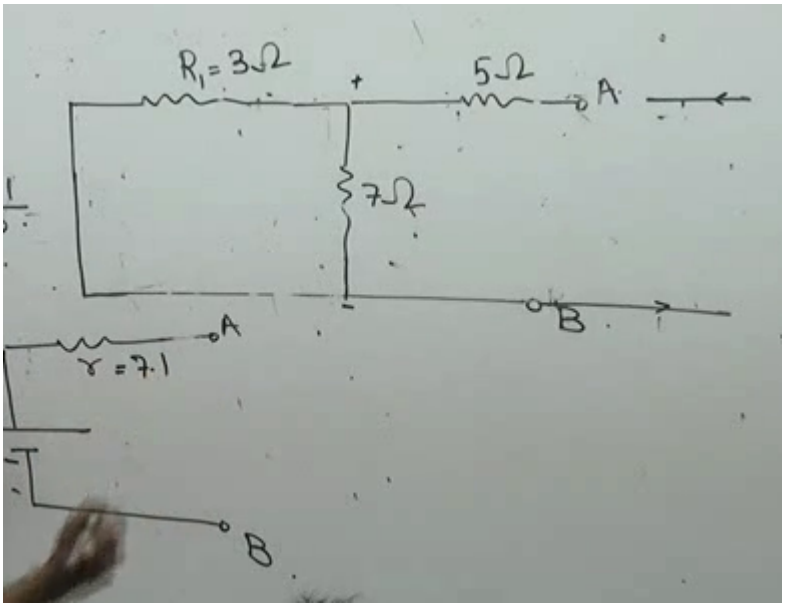
DC Circuit Diagram (Completed state, 23:13)



Electrical circuit (Initial state, 23:20)

Calculating Thevenin Voltage (EPSA)

To find the Thevenin voltage, the circuit is opened across terminals A and B (no load). The potential difference between A and B is then calculated. Given the circuit configuration, the current flowing through the series combination of the 3 Ω and 7 Ω resistors is calculated as $I = \frac{10\,\text{V}}{3\,\Omega + 7\,\Omega} = 1\,\text{A}$. The voltage drop across the 7 Ω resistor is $V = I \times R = 1\,\text{A} \times 7\,\Omega = 7\,\text{V}$. Since there is no current flowing through the 5 Ω resistor in the open-circuit state, the potential difference between A and B is equal to the voltage drop across the 7 Ω resistor, resulting in an EPSA of 7 V.



Electrical Circuit Diagram (Completed state, 25:55)

Calculating Thevenin Resistance (r)

To find the Thevenin resistance (r), all voltage sources are short-circuited. With the 10 V source shorted, the 3 Ω and 7 Ω resistors are in parallel. The equivalent resistance of this parallel combination is $\frac{3\times7}{3+7} = \frac{21}{10} = 2.1\,\Omega$. This combination is in series with the 5 Ω resistor. Therefore, the total Thevenin resistance is $r = 5\,\Omega + 2.1\,\Omega = 7.1\,\Omega$.

Final Equivalent Circuit and Load Current

The circuit is reduced to a 7 V voltage source in series with a 7.1 Ω resistance. For any load R connected across A and B, the current I_0 through the load is given by the formula:

$$I_0 = \frac{7}{7.1 + R}$$

Maximum Power Transfer Theorem Application

Problem Setup: Determining Load Resistance for Maximum Power

The circuit is characterized by an internal resistance $\gamma = 5 + \frac{21}{10} = 7.1\,\Omega$ and an electromotive force $\epsilon = 7\,\text{V}$. The current i_0 is given by the expression $i_0 = \frac{7}{7.1+R}$,

where R is the load resistance connected between terminals A and B.

Applying the Maximum Power Transfer Theorem

To achieve maximum power dissipation across the load resistance R, the load resistance must be equal to the internal resistance of the source. Given the internal resistance $\gamma = 7.1\,\Omega$, the required load resistance R is $7.1\,\Omega$.